Piecewise-Deterministic Markov Processes

Theory and Methods for Stationary Distributions

Andres Ferragut

Universidad ORT Uruguay

Outline

Motivation and Context

Definition of the PDMP

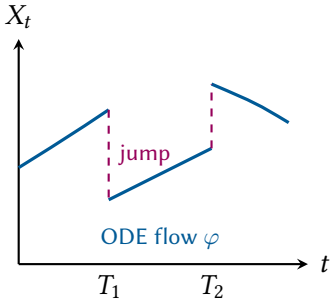
Forward Kolmogorov Equation

Stability, ergodicity

Piecewise deterministic Markov processes

Davis (1984) observed that continuous-time stochastic models consist of:

1. **Diffusion** (Brownian noise)
2. **Deterministic motion**
3. **Random jumps**



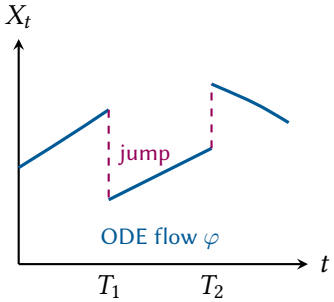
The Gap

Diffusion theory is highly unified (Itô calculus). Non-diffusion models — ingredients (2)+(3) — were a *heterogeneous collection of special cases.*

Piecewise deterministic Markov processes

Davis (1984) observed that continuous-time stochastic models consist of:

1. **Diffusion** (Brownian noise)
2. **Deterministic motion**
3. **Random jumps**



The Gap

Diffusion theory is highly unified (Itô calculus). Non-diffusion models — ingredients (2)+(3) — were a *heterogeneous collection of special cases*.

Goal: Provide a *canonical class* covering virtually all non-diffusion models, with a unified stochastic calculus analogous to diffusion theory.

Motivating Examples

Classical examples (Davis 1984)

- Countable-state Markov processes
- Non-exponential sojourn-time processes
- $M/G/1$ and $GI/G/1$ queues (virtual waiting time)
- Dam / storage / release models

Modern applications

- TCP/IP congestion control
- Reliability & maintenance models
- Biochemical reaction networks
- Risk / ruin theory (insurance)

Key claim (Davis 1984)

The class is *"wide enough to include as special cases virtually all the non-diffusion models of applied probability."*

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(1) The flow φ

Unique solution map of $\dot{x} = f(x)$, so

$$\varphi(0, x) = x$$

$$\varphi(t+s, x) = \varphi(t, \varphi(s, x))$$

Assumes f is Lipschitz continuous.

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(2) The jump rate function λ

Measurable $\lambda : E \to \mathbb{R}_+$.

Integrated rate:

$$\Lambda(t, x) = \int_0^t \lambda(\varphi(s, x))\, \mathrm{d}s$$

governs the Poisson-like clock.

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(3) The jump kernel Q

Probability measure $Q(x, \cdot)$ on E.

Post-jump location

$$Z_1 \sim Q(\varphi(T_1, x_0), \cdot)$$

drawn independently of T_1.

The Integrated Hazard Rate Λ

The function $\lambda : E \to \mathbb{R}_+$ is a **hazard rate**: it measures the instantaneous intensity of jumping *at the current location of the flow.*

Because the process moves deterministically between jumps, the relevant object is the **integrated hazard along the trajectory**:

$$\Lambda(t, x) \;=\; \int_0^t \lambda\big(\varphi(s, x)\big)\, \mathrm{d}s.$$

This is the cumulative intensity seen by a particle starting at x and following the ODE flow for time t.

The first jump time T_1 is then defined as a *non-homogeneous exponential* with this integrated rate:

$$P(T_1 > t \mid X_0 = x) \;=\; e^{-\Lambda(t,\, x)}.$$

Boundary condition

- **Reachable boundary:**
  $\partial^* E = \{z \in \partial E : \varphi(-t, z) \in E \text{ for all } t \in (0, \varepsilon), \text{ some } \varepsilon > 0\}$ — boundary points that the flow can reach from inside E.
- **Boundary exit time:** $t^*(x) = \inf\{t > 0 : \varphi(t, x) \in \partial^* E\}$, with $t^*(x) = \infty$ if the flow never reaches $\partial^* E$.

Davis's Assumption (boundary jumps)

If the flow reaches the boundary in finite time, i.e. $t^*(x) < \infty$, Davis assumes

$$\Lambda(t^*(x),\, x) = \infty.$$

Consequence: $P(T_1 < t^*(x)) = 1$, so the Poisson clock *always fires before the boundary is reached*. Forced boundary jumps are excluded.

Remark: Costa (1990) relaxes this assumption.

Sample Path Construction

Starting from $x_0 \in E$, one inter-jump cycle is constructed as follows.

1. **Draw the alarm level.** Sample $U \sim \mathrm{Uniform}(0,1)$ and set $s^* = -\log U > 0$.
2. **Run the flow and clock jointly.** Integrate $\dot{x} = f(x)$ from x_0 while simultaneously accumulating $\Lambda(t, x_0) = \int_0^t \lambda(\varphi(s, x_0))\,\mathrm{d}s$. Stop at the first time

$$T_1 = \inf\left\{t > 0 : \Lambda(t, x_0) \geq s^*\right\}.$$

   Under Davis's assumption, $T_1 < t^*(x_0)$ with probability 1.
3. **Draw the post-jump location.** Sample $Z_1 \sim Q\left(\varphi(T_1, x_0), \cdot\right)$ independently of T_1.
4. **Restart.** Set $X_{T_1} = Z_1$ and repeat from step 1.

Markov Property and Non-Explosion

Markov property

The semigroup property of φ ensures the process is **strong Markov**: the distribution of the next jump time depends only on the current state X_t, not on the history.

Non-explosion (Davis, Assumption 3.1)

Require $E[N_t] < \infty$ for all $t \geq 0$, equivalently $T_n \to \infty$ a.s., so the process does not accumulate infinitely many jumps in finite time.

Remark

Non-explosion is not automatic — it must be verified for each model.

Example: TCP/IP Congestion Control

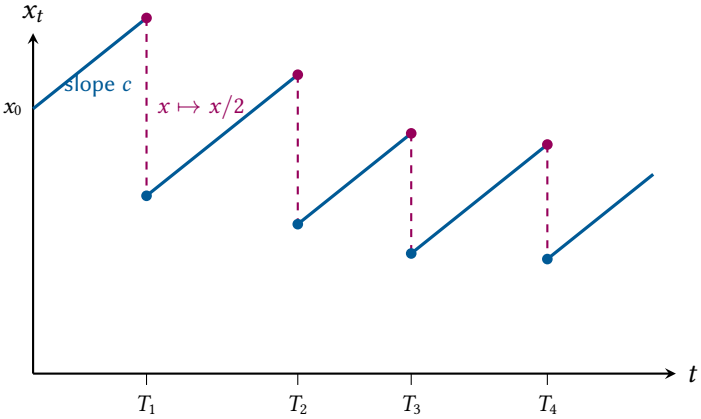
Informal description

A TCP flow increases its sending rate continuously (deterministic motion) and suffers random packet loss events (jumps) that halve the rate — the classical AIMD mechanism.

Local characteristics:

- **State space:** $E = \mathbb{R}_+$, where x is the current sending rate.
- **Flow:** $f(x) = c > 0$ (additive increase), so $\varphi(t, x) = x + ct$.
- **Jump rate:** $\lambda(x) = \alpha x$, $\alpha > 0$ — loss probability grows with the sending rate.
- **Jump kernel:** $Q(x, \cdot) = \delta_{x/2}$ — upon a loss event the rate is halved.

Example: TCP/IP Congestion Control



The Generator: Motivation

We want to characterize how $E[\psi(X_t)]$ evolves. For a smooth test function $\psi : E \to \mathbb{R}$, track $\psi(X_t)$ along a sample path.

Between jumps: $X_t = \varphi(t, x)$ follows the ODE, so by the chain rule:

$$\frac{d}{dt}\psi(\varphi(t,x)) = f(\varphi(t,x)) \cdot \nabla\psi(\varphi(t,x)).$$

The instantaneous rate of change is the **directional derivative of ψ along the vector field f.**

At a jump time T_i: the process jumps from $X_{T_i^-}$ to $Z_i \sim Q(X_{T_i^-}, \cdot)$, so ψ changes by:

$$\psi(Z_i) - \psi(X_{T_i^-}).$$

Taking expectations, the average jump contribution at x is:

$$\lambda(x) \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y).$$

The Generator: Domain $\mathcal{D}(\mathcal{A})$

Not every function ψ can be in the domain of $\mathcal{A}$. Davis identifies three conditions.

1. **Absolute continuity along trajectories.** $t \mapsto \psi(\varphi(t,x))$ is absolutely continuous on $[0, t^*(x))$ for each $x \in E$.
   Why: ensures the directional derivative $f(x) \cdot \nabla\psi(x)$ exists Lebesgue-a.e. along each trajectory, so the ODE contribution is well defined.

2. **Boundary condition.** $\psi(x) = \int_E \psi(y)\, Q(x, \mathrm{d}y)$ for all $x \in \partial^* E$.
   Why: relevant when Davis's assumption is relaxed and forced boundary jumps can occur (Costa 1990). Under Davis's assumption this condition is vacuous.

3. **Integrability.** $\mathcal{B}\psi \in L^{1,\mathrm{loc}}(p)$, where $\mathcal{B}\psi(x) = \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y)$.
   Why: ensures the stochastic integral against the jump martingale measure is well defined.

The Extended Generator: Davis (1984) Theorem 5.5

Theorem (Davis 1984, Theorem 5.5)

For $\psi \in \mathcal{D}(\mathcal{A})$, the extended generator of the PDMP is:

$$\boxed{\mathcal{A}\psi(x) \;=\; \underbrace{f(x)\cdot\nabla\psi(x)}_{\textit{ODE drift}} \;+\; \underbrace{\lambda(x)\int_E \big[\psi(y)-\psi(x)\big]\,Q(x,\mathrm{d}y)}_{\textit{average jump contribution}}}$$

Martingale connection

Dynkin formula

For $\psi \in \mathcal{D}(\mathcal{A})$, the process

$$C_t^{\psi} \;=\; \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{A}\psi(X_s)\,\mathrm{d}s$$

is a local $\mathcal{F}_t$-martingale. This is the fundamental connection between the generator and the process.

- $\mathcal{A}$ is not a local operator — the integral term couples $\psi(x)$ to all states y reachable by a jump.
- Unlike diffusion, the calculus here is *ordinary* calculus with a stochastic interpretation — no Itô correction terms.

Kolmogorov Forward Equation: Heuristic

Let $p(x,t)$ denote the density of X_t. Track the probability mass in a small region $A \subset E$ over a small time interval $[t, t+dt]$:

- **Loss due to flow.** The ODE carries mass out of A through its boundary at rate $f(x) \cdot n(x)$ where $n(x)$ is the outward normal. By the divergence theorem:

$$\text{loss}_{\text{flow}} = \int_A \nabla \cdot [f(x)\, p(x,t)]\, \mathrm{d}x.$$

- **Loss due to jumps.** Mass at $x \in A$ jumps away at rate $\lambda(x)$:

$$\text{loss}_{\text{jump}} = \int_A \lambda(x)\, p(x,t)\, \mathrm{d}x.$$

- **Gain due to jumps.** Mass at $y \notin A$ jumps into A with probability $Q(y,A)$ at rate $\lambda(y)$:

$$\text{gain}_{\text{jump}} = \int_E \lambda(y)\, p(y,t)\, Q(y,A)\, \mathrm{d}y = \int_A \int_E \lambda(y)\, p(y,t)\, Q(y,\mathrm{d}z)\, \mathrm{d}x.$$

Kolmogorov Forward Equation: Formal Derivation

Starting from the Dynkin formula, for any $\psi \in \mathcal{D}(\mathcal{A})$:

$$\frac{d}{dt}E[\psi(X_t)] = E[\mathcal{A}\psi(X_t)] = \int_E \mathcal{A}\psi(x)\, p(x,t)\, \mathrm{d}x.$$

But also $\frac{d}{dt}E[\psi(X_t)] = \int_E \psi(x)\, \partial_t p(x,t)\, \mathrm{d}x$. So we need the L^2-adjoint $\mathcal{A}^*$ satisfying $\int \psi\, \mathcal{A}^* p = \int \mathcal{A}\psi \cdot p$ for all ψ.

ODE term. Integration by parts (no boundary terms under Davis's assumption):

$$\int_E f(x) \cdot \nabla\psi(x)\, p(x,t)\, \mathrm{d}x = -\int_E \psi(x)\, \nabla \cdot [f(x)\, p(x,t)]\, \mathrm{d}x.$$

Jump term. Expanding and swapping the role of x and y in the gain part:

$$\int_E \lambda(x) \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y)\, p(x,t)\, \mathrm{d}x$$
$$= \int_E \psi(x) \left[\int_E \lambda(y)\, p(y,t)\, Q(y, \mathrm{d}x) - \lambda(x)\, p(x,t) \right] \mathrm{d}x.$$

Kolmogorov Forward Equation: Result

Theorem (Kolmogorov Forward Equation)

Assuming X_t has density $p(x,t)$, it satisfies:

$$\boxed{\partial_t p(x,t) = \underbrace{-\nabla \cdot [f(x)\, p(x,t)]}_{\textit{transport}} \underbrace{- \lambda(x)\, p(x,t)}_{\textit{loss via jumps}} + \underbrace{\int_E \lambda(y)\, p(y,t)\, Q(y, \mathrm{d}x)}_{\textit{gain via jumps}}}$$

Transport term

First-order hyperbolic PDE. No diffusion. Solved by method of characteristics along $\dot{x} = f(x)$.

Loss term

Probability mass leaves x at rate $\lambda(x)$. Local operator.

Gain term

Non-local integral operator. Couples the density at all states y jumping into x.

Stationary Condition

Setting $\partial_t p = 0$ in the Kolmogorov forward equation, a density $\pi(x)$ is stationary if and only if:

$$\boxed{\nabla \cdot [f(x)\,\pi(x)] + \lambda(x)\,\pi(x) = \int_E \lambda(y)\,\pi(y)\,Q(y, \mathrm{d}x)}$$

- The left-hand side is a local operator: transport out of x plus probability mass lost via jumps at x.
- The right-hand side is non-local: probability mass arriving at x from jumps originating at all $y \in E$.
- This is an integro-differential equation — generally hard to solve analytically.

Ergodicity

Definition

A PDMP with semigroup (P_t) is **ergodic** if it admits a unique invariant probability measure π, satisfying

$$\int_E P_t\psi \, \mathrm{d}\pi = \int_E \psi \, \mathrm{d}\pi \quad \text{for all bounded measurable } \psi, \ t \geq 0,$$

and for all initial conditions $x \in E$:

$$\mu_t \xrightarrow{t\to\infty} \pi \quad \text{weakly},$$

where $\mu_t = P_t(x, \cdot)$ is the distribution of X_t starting from $X_0 = x$.

Ergodicity

Stronger notions

- **Positive recurrence:** the process returns to recurrent sets in finite expected time (defined precisely after petite sets are introduced).

- **Geometric ergodicity:** convergence is exponentially fast in a weighted norm $\|\cdot\|_V$ for some $V \geq 1$.

The Ergodic Theorem

Theorem (Ergodic theorem, Meyn & Tweedie (1993) Theorem 13.0.1)

Let the PDMP be ergodic with invariant measure π. Then for all $x \in E$ and all $g \in L^1(\pi)$:

$$\frac{1}{T}\int_0^T g(X_t)\,\mathrm{d}t \xrightarrow{T\to\infty} \int_E g\,\mathrm{d}\pi \quad a.s.$$

No rate without more structure

Ergodicity gives convergence but no rate. Positive recurrence and geometric ergodicity provide quantitative control on how fast $\mu_t \to \pi$.

Irreducibility: A Necessary Condition

Definition (ν-irreducibility, Meyn & Tweedie (1993) Chapter 4)

A PDMP is ν-**irreducible** with respect to a σ-finite measure ν on E if for every $x \in E$ and every measurable A with $\nu(A) > 0$:

$$\int_0^\infty P_t(x, A)\,\mathrm{d}t > 0.$$

What this requires for a PDMP

- **ODE reachability.** From x, the flow $\varphi(t, x)$ must reach a neighborhood of some point y from which a jump into A is possible.
- **Jump support.** $Q(y, A) > 0$ for some y reachable by the flow from x.

Irreducibility: Necessary but Not Sufficient

Irreducibility is necessary but not sufficient

Without irreducibility, ergodicity cannot hold: the process could be confined to a strict subset of E forever. However, irreducibility alone gives no information about convergence — additional conditions are needed.

When irreducibility fails: limit cycles

If the ODE has an invariant set $\mathcal{L} \subset E$ (e.g. a limit cycle) and $Q(x, \mathcal{L}) = 1$ for all $x \in \mathcal{L}$, then the process started on $\mathcal{L}$ can never leave it:

$$P_x(X_t \in \mathcal{L} \text{ for all } t \geq 0) = 1, \quad x \in \mathcal{L}.$$

Any A disjoint from $\mathcal{L}$ satisfies $\int_0^\infty P_t(x, A)\,\mathrm{d}t = 0$, so ν-irreducibility fails.

Harris Recurrence: Returning Infinitely Often

Irreducibility guarantees the process can reach any set, but not that it returns *infinitely often*.

Definition (Harris recurrence, Meyn & Tweedie (1993) Chapter 9)

A ν-irreducible PDMP is **Harris recurrent** if for every measurable A with $\nu(A) > 0$:

$$P_x\left(\int_0^\infty \mathbf{1}_{X_t \in A}\,\mathrm{d}t = \infty\right) = 1 \quad \text{for all } x \in E.$$

Why Harris recurrence is needed

Without it, the process could visit every set at least once but then drift away to infinity, never to return. Harris recurrence rules this out: every positive-measure set is visited infinitely often with probability 1.

Harris Recurrence: Connection to Ergodicity

Harris Ergodic Theorem,Meyn & Tweedie (1993), Theorem 13.0.1.

A ν-irreducible, Harris recurrent, positively recurrent PDMP admits a unique invariant measure π and satisfies

$$\mu_t \xrightarrow{t\to\infty} \pi \quad \text{weakly}, \quad \text{for } \textit{all } x \in E.$$

The convergence holds for *every* starting point $x \in E$, not just π-almost every one. This is the key advantage of Harris recurrence over standard ergodicity: it gives a *uniform* guarantee across all initial conditions.

What is still missing

Positive recurrence has not yet been defined — it requires the notion of a **petite set**, introduced next.

Petite Sets: Definition

Definition (Petite set, Meyn & Tweedie (1993) Chapter 5)

A measurable set $C \subset E$ is **petite** if there exist $T > 0$, $\varepsilon > 0$ and a probability measure ν_C on E such that:

$$P^T(x, \cdot) \geq \varepsilon \, \nu_C(\cdot) \quad \text{for all } x \in C.$$

Intuition: the refreshment mechanism

Every visit to C carries a probability ε of *refreshing* the distribution to ν_C, regardless of history. This breaks dependence on the initial condition and forces convergence to π.

Without petite sets, two trajectories starting at different points could both visit C infinitely often yet never couple — their distributions would never mix.

Petite Sets: Verification for PDMPs

Sufficient conditions for a compact set C to be petite, Costa & Dufour (1999) Section 3

- **Jump rate bounded below:** $\lambda(x) \geq \lambda_{\min} > 0$ for all $x \in C$. Ensures jumps occur at a positive rate throughout C.
- **Jump kernel has a density bounded below:** $Q(x, \cdot)$ has a density bounded away from zero on C. Ensures jumps spread mass uniformly.
- **Local ODE controllability:** the flow connects points in C, so the process can reach any part of C from any other part.

Positive Recurrence

Definition (Positive recurrence, Meyn & Tweedie (1993) Theorem 10.0.1)

A Harris recurrent PDMP is **positively recurrent** if for some petite set C with $\nu(C) > 0$:

$$E_x[\tau_C] < \infty \quad \text{for all } x \in E,$$

where $\tau_C = \inf\{t > 0 : X_t \in C\}$.

Independence of the choice of C

If $E_x[\tau_C] < \infty$ holds for one petite set C, it holds for *all* petite sets with $\nu(C) > 0$. The definition is therefore well-posed.

Positive Recurrence: Connection to π

Invariant measure via excursions, Meyn & Tweedie (1993) Theorem 10.0.1

Positive recurrence is equivalent to the existence of a finite invariant measure π. Moreover:

$$\pi(A) \propto E_x\left[\int_0^{\tau_C} \mathbf{1}_{X_t \in A}\, \mathrm{d}t\right],$$

i.e. $\pi(A)$ is proportional to the expected time spent in A during one excursion from C.

Interpretation

The invariant measure π is built from the local behaviour of the process during a single return cycle to C. This gives a constructive characterisation of π that does not require solving the stationary equation directly.

Summary so far

We have shown that:

$$\nu\text{-irreducibility} + \text{Harris recurrence} + \text{positive recurrence} \implies \text{ergodicity}.$$

The next slides give *sufficient conditions* for all three to hold simultaneously.

A Sufficient Condition for Ergodicity: Foster's Criterion

Theorem (Foster's criterion, Meyn & Tweedie (1993) Theorem 3.2; Costa & Dufour (1999) Theorem 3.1)

Suppose the PDMP is ν-irreducible. If there exist $V \geq 1$, a petite set C, and constants $\varepsilon > 0$, $b < \infty$ such that:

$$\mathcal{A}V(x) \leq -\varepsilon + b\,\mathbf{1}_C(x),$$

then the PDMP is Harris recurrent, positively recurrent, and admits a unique invariant distribution π.

Interpretation: Outside C, the expected infinitesimal change in V is strictly negative — the process is pushed back towards C at a uniform rate ε. Inside C, V may increase by at most b. The Lyapunov function V measures how far the process is from the petite set C.

Foster's Criterion: Bound on Return Times

Derivation via the Dynkin formula

Apply the Dynkin formula to V up to τ_C. Since $C_t^V = V(X_t) - V(X_0) - \int_0^t \mathcal{A}V(X_s)\,\mathrm{d}s$ is a local martingale, taking expectations and using optional stopping at τ_C:

$$E_x[V(X_{\tau_C})] - V(x) = E_x\left[\int_0^{\tau_C} \mathcal{A}V(X_s)\,\mathrm{d}s\right] \leq -\varepsilon\, E_x[\tau_C] + b.$$

Since $V \geq 1$ and $E_x[V(X_{\tau_C})] \geq 1$:

$$\varepsilon\, E_x[\tau_C] \leq V(x) - 1 + b,$$

so:

$$E_x[\tau_C] \leq \frac{V(x) - 1 + b}{\varepsilon} < \infty \quad \text{for all } x \in E.$$

Consequence

A Sufficient Condition for Geometric Ergodicity

Theorem (Meyn & Tweedie (1993), Durmus, Guillin & Monmarché (2021))

Suppose the PDMP is ν-irreducible. If there exist $V \geq 1$, a petite set C, and constants $c > 0$, $b < \infty$ such that:

$$\mathcal{A}V(x) \leq -c\, V(x) + b\, \mathbf{1}_C(x),$$

then the PDMP is geometrically ergodic: there exist $C_0 < \infty$ and $\varrho > 0$ such that

$$\|\mu_t - \pi\|_V \leq C_0\, V(x_0)\, e^{-\varrho t}.$$

Strengthening of Foster's criterion

The $-cV$ drift forces faster returns to C for states with large V: the further the process is from C, the stronger the pull back. This quantitative control on return times delivers exponential convergence.

TCP/IP Congestion Control: The Model

Local characteristics

- **State space:** $E = \mathbb{R}_+$, where x is the current sending rate. No reachable boundary: $t^*(x) = \infty$ for all $x \in E$.
- **Flow:** $f(x) = c > 0$ (additive increase), so $\varphi(t, x) = x + ct$.
- **Jump rate:** $\lambda(x) = \alpha x$, $\alpha > 0$ — loss probability grows with the sending rate.
- **Jump kernel:** $Q(x, \cdot) = \delta_{x/2}$ — upon a loss event the rate is halved.

Generator

For $\psi \in \mathcal{D}(\mathcal{A})$:

$$\mathcal{A}\psi(x) = c\,\psi'(x) + \alpha x \left[\psi\left(\tfrac{x}{2}\right) - \psi(x)\right]$$

TCP/IP: Irreducibility

Claim

The TCP/IP PDMP is ν-irreducible with respect to the Lebesgue measure ν on $\mathbb{R}_+$.

Argument. Fix $x > 0$ and any interval $(a, b) \subset \mathbb{R}_+$.

- **ODE reachability.** The flow $\varphi(t, x) = x + ct$ is strictly increasing, so from x the process can reach any value $y > x$ in finite time $t = (y - x)/c$.
- **After one jump.** From y, a jump sends the process to $y/2$. By choosing $y = 2z$ for any $z \in (a, b)$, the process lands in (a, b) after one jump.
- **For $x > 2b$.** First flow to $y = 2z \in (2a, 2b)$, then jump to $z \in (a, b)$. Since $\lambda(y) = \alpha y > 0$, this occurs with positive probability.

Hence $\int_0^\infty P_t(x, (a, b))\, \mathrm{d}t > 0$ for all $x > 0$ and all (a, b), so ν-irreducibility holds.

TCP/IP: Petite Set

Claim

For any $0 < a < b < \infty$, the compact set $C = [a, b]$ is petite.

Argument. We need to find $T > 0$, $\varepsilon > 0$ and ν_C such that $P^T(x, \cdot) \geq \varepsilon\, \nu_C(\cdot)$ for all $x \in C$.
Since $Q(x, \cdot) = \delta_{x/2}$ is a point mass, we argue via **two jumps**:

- After the first jump from $x \in [a, b]$, the process lands at $x/2 \in [a/2, b/2]$ and then evolves under the flow.
- After a second jump from some $y \in [a/2 + ct, b/2 + ct]$, the process lands at $y/2$, which has an **absolutely continuous** distribution since y varies continuously with the flow time t.
- On C: $\lambda(x) \geq \alpha a > 0$, so both jumps occur at a positive rate.

After two jumps, the distribution of X has an absolutely continuous component bounded below by some $\varepsilon\, \nu_C$ on a compact set. Hence C is petite.

TCP/IP: Ergodicity via Foster's Criterion

Lyapunov function: $V(x) = x$

$$\mathcal{A}V(x) = c \cdot 1 + \alpha x \left[\frac{x}{2} - x\right] = c - \frac{\alpha x^2}{2}$$

Applying Foster's criterion. Choose $\delta > 0$ small and set $C = \left[\delta, \sqrt{\frac{2c}{\alpha}} + \delta\right]$. Then:

- **Outside** C, $x > \sqrt{2c/\alpha} + \delta$:

$$\mathcal{A}V(x) = c - \frac{\alpha x^2}{2} \leq c - \frac{\alpha}{2}\left(\sqrt{\frac{2c}{\alpha}} + \delta\right)^2 = -\alpha\delta\sqrt{\frac{c}{2\alpha}} - \frac{\alpha\delta^2}{2} =: -\varepsilon < 0.$$

- **Inside** C: $\mathcal{A}V(x) \leq c =: b < \infty$.

Hence $\mathcal{A}V(x) \leq -\varepsilon + b\,\mathbf{1}_C(x)$ with $\varepsilon > 0$ and $b = c < \infty$.

TCP/IP: Ergodicity via Foster's Criterion

Conclusion:

Since the PDMP is ν-irreducible and C is a petite set, Foster's criterion (Meyn & Tweedie 1993, Theorem 3.2) gives Harris recurrence, positive recurrence, and existence of a unique stationary distribution π.

TCP/IP: Stationary Forward Equation

The stationary Kolmogorov forward equation for the density $\pi(x)$ is:

$$\boxed{c\,\pi'(x) + \alpha x\,\pi(x) = 4\alpha x\,\pi(2x)}$$

Derivation of the right-hand side. The gain term is:

$$\int_0^\infty \lambda(y)\,\pi(y)\,Q(y,\mathrm{d}x) = \int_0^\infty \alpha y\,\pi(y)\,\delta_{y/2}(\mathrm{d}x)\,\mathrm{d}y.$$

The measure $\delta_{y/2}(\mathrm{d}x)$ concentrates at $y = 2x$, so changing variables $y = 2x$, $\mathrm{d}y = 2\,\mathrm{d}x$:

$$= \alpha \cdot 2x \cdot \pi(2x) \cdot 2 = 4\alpha x\,\pi(2x).$$

Non-locality

The equation couples $\pi(x)$ at scale x to $\pi(2x)$ at scale $2x$ — a **functional differential equation**. Standard ODE methods and method of characteristics do not apply directly. The natural tool is the **Mellin transform** (Baccelli, 2002).

TCP/IP: Second Moment in Steady State

Since the PDMP is ergodic with stationary distribution π, for any $\psi \in \mathcal{D}(\mathcal{A})$:

$$\int_E \mathcal{A}\psi \, \mathrm{d}\pi = 0.$$

Choose $\psi(x) = x$. Recall:

$$\mathcal{A}\psi(x) = c - \frac{\alpha x^2}{2}.$$

Applying $\int \mathcal{A}\psi \, \mathrm{d}\pi = 0$:

$$c - \frac{\alpha}{2} E_\pi[X^2] = 0 \implies \boxed{E_\pi[X^2] = \frac{2c}{\alpha}.}$$

Mathis formula revisited...

Bound on $E_\pi[X]$ via Jensen's inequality

Since $x \mapsto x^2$ is convex, Jensen's inequality gives:

$$E_\pi[X]^2 \leq E_\pi[X^2] = \frac{2c}{\alpha},$$

hence:

$$E_\pi[X] \leq \sqrt{\frac{2c}{\alpha}}.$$

Remark: The exact value of $E_\pi[X]$ requires more advanced techniques (Mellin transform).

Summary

Foundations (Davis 1984)

- PDMPs: deterministic flow punctuated by random jumps, characterised by (φ, λ, Q).
- Extended generator $\mathcal{A}$: ODE drift + non-local jump term.
- Kolmogorov forward equation: hyperbolic PDE + integral operator.
- Dynkin formula: bridge between $\mathcal{A}$ and martingales.

Ergodicity

- Irreducibility + Harris recurrence + positive recurrence $\implies$ ergodicity.
- Foster's criterion: sufficient condition via $\mathcal{A}V \leq -\varepsilon + b\mathbf{1}_C$.
- Lyapunov criterion: geometric ergodicity via $\mathcal{A}V \leq -cV + b\mathbf{1}_C$.

TCP/IP example

- Ergodicity verified via Foster's criterion.
- $E_\pi[X^2] = 2c/\alpha$ from the generator.
- $E_\pi[X] \leq \sqrt{2c/\alpha}$ via Jensen.

References I

- M. H. A. Davis. *Piecewise-deterministic Markov processes: A general class of non-diffusion stochastic models.* J. R. Statist. Soc. B **46**(3):353–388, 1984.
- M. H. A. Davis. *Markov Models and Optimization.* Chapman & Hall, 1993.
- O. L. V. Costa. *Stationary distributions for piecewise-deterministic Markov processes.* J. Appl. Prob. **27**:60–73, 1990.
- R. L. Tweedie. *Sufficient conditions for ergodicity and recurrence of Markov chains on a general state space.* Stoch. Proc. Appl. **3**:385–403, 1975.
- D. Vermes. *Optimal control of piecewise deterministic processes.* Stochastics **14**:165–208, 1985.
- F. Dufour, O. L. V. Costa. *Stability of piecewise deterministic Markov processes.* SIAM J. Control Optim. **37**(5):1483–1502, 1999.
- A. Durmus, A. Guillin, P. Monmarche. *Piecewise deterministic Markov processes and their invariant measures.* Ann. Inst. H. Poincaré **57**(3), 2021.

References II

- S. P. Meyn, R. L. Tweedie. *Markov Chains and Stochastic Stability.* Springer-Verlag, London, 1993. (2nd edition: Cambridge University Press, 2009.)
- S. P. Meyn, R. L. Tweedie. *Stability of Markovian processes III: Foster–Lyapunov criteria for continuous-time processes.* Adv. Appl. Prob. **25**(3):518–548, 1993.
- F. Baccelli, D. McDonald, J. Reynier. *A mean-field model for multiple TCP connections through a buffer implementing RED.* Performance Evaluation **49**(1–4):77–97, 2002.